

ASSIGNMENT 0 — COMPUTER LANGUAGES

COMP 3010 — ORGANIZATION OF PROGRAMMING LANGUAGES

1. COMPILERS AND INTERPRETERS

Exercise 1 (Short Answer). In your own words, explain the task of each of the following phases of a compiler:

- (1) Lexical analyzer (*a.k.a.* lexer)
- (2) Syntax analyzer (*a.k.a.* parser)
- (3) Semantic analyzer
- (4) Intermediate code generator
- (5) Code optimizer
- (6) Machine code generator

Exercise 2 (Short Answer). In your own words, explain the difference between a compiler and an interpreter.

Exercise 3 (This or That). Which of the following are advantages of compilers? Which are advantages of interpreters? *Explain your choice in each part.*

- (1) Can pre-examine input program for semantic (*e.g.*, type) errors
- (2) Have full knowledge of both program input and program implementation
- (3) Can specialize parts of code to optimize for specific input data
- (4) Can afford heavy weight optimizations over large sections of code
- (5) Flexible, and can easily change program behavior dynamically at run-time
- (6) Generated code can run many times

2. COMPUTABILITY

Exercise 4 (Short Answer). (1) If Q is a program that solves the halting problem, what must $Q(P, x)$ do? What kind of answer does it return, and what does that mean about the arguments P (representing a program) and x (representing input to P)?

- (2) What is the significance of the halting problem for computable functions? What ramifications does it have for what we can do?

Exercise 5 (Short Answer). Suppose that the program Z is a *zero checker*. That is, Z takes two inputs, a string P representing a program and a number n , and checks whether or not running P with input n (written as $P(n)$) would output a 0:

$$Z(P, n) = \begin{cases} \text{true} & \text{if } P(n) \text{ returns } 0 \\ \text{false} & \text{otherwise} \end{cases}$$

Because the halting problem is uncomputable, we know that the *zero checker* is not a computable function. Show why the program Z cannot exist by explaining how it could be used to solve the halting problem.